Abstract Algebra

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1 Abstract Algebra (抽象代數)

I Operation (運算)

An n-ary operation on a set A is a function from a subset of A^n to A.

II Algebraic structures (代數結構) or algebraic system (代數系統)

An algebraic structure or algebraic system consists of a nonempty set A, called the underlying set, carrier (set), or domain, a collection of operations on A, and a finite set of identities called axioms that these operations must satisfy.

III Common axioms

i Equational axioms

- Commutativity (交換律): An operation $*$ is commutative if $x * y = y * x$ for every x and y in the algebraic structure.
- Associativity (結合律): An operation $*$ is associative if $(x * y) * z = x * (y * z)$ for every x, y and z in the algebraic structure.
- Left distributivity (左分配律): An operation $*$ is left-distributive with respect to another operation $+$ if $x * (y + z) = (x * y) + (x * z)$ for every x, y, and z in the algebraic structure.
- Right distributivity (右分配律): An operation $*$ is right-distributive with respect to another operation $+$ if $(y+z)*x = (y*x)+(z*x)$ for every x, y, and z in the algebraic structure.
- Distributivity (分配律): An operation $*$ is distributive with respect to another operation $+$ if it is both left-distributive and right-distributive with respect to $+$. If the operation $*$ is commutative, left and right distributivity are both equivalent to distributivity.

ii Existential axioms

- Having an identity element: A binary operation $*$ has an identity element if there is an element e such that $x * e = x$ and $e * x = x$ for any x in the algebraic structure. Unless otherwise specified, the identity element of an operator denoted as $+$ is denoted as 0, and the identity element of an operator denoted as $*$ is denoted as 1.
- Latin square property: A binary operation $*$ has Latin square property iff for every a, b in the algebraic structure, there exists two unique x, y in the algebraic structure such that $a * x = b$ and $x * a = b$.

- Invertibility: Given a binary operation $*$ that an identity element e, an element x in the algebraic structure is invertible if there exists an unique element $\text{inv}(x)$, called the inverse element, in the algebraic structure such that $x * \text{inv}(x) = e$ and $\text{inv}(x) * x = e$. Unless otherwise specified, the inverse element of a of an operator denoted as $+$ is denoted as $-a$, and the inverse element of a of an operator denoted as $*$ is denoted as a^{-1}.

IV Group-like algebraic structure

i Magma (原群) or binar

A magma or binar is a set M with a binary operation $*$ such that

$$\forall a, b : \ a, b \in M \implies a * b \in M.$$

ii Semigroup (半群)

A semigroup is a magma equipped with a binary operation that is associative.

iii Unital magma

A unital magma is a magma equipped with a binary operation that has an identity element.

iv Monoid (么半群, 單群, or 亞群)

A monoid is a magma equipped with a binary operation that is associative and has an identity element.

v Commutative monoid or abelian monoid

A commutative monoid or abelian monoid is a monoid equipped with a binary operation that is commutative.

vi Quasigroup (擬群)

A quasigroup is a magma equipped with a binary operation that has Latin square property.

vii Loop (么擬群 or 圈)

A loop is a magma equipped with a binary operation that has Latin square property and has an identity element.

viii Associative quasigroup

An associative quasigroup is a magma equipped with a binary operation that is associative and has Latin square property.

ix Group (群)

A group is a magma equipped with a binary operation that has an identity element, is associative, and every element has an inverse element, that is, a nonempty associative quasigroup.

x Abelian group (阿貝爾群) or commutative group (交換群)

An abelian group, also called a commutative group, is a group equipped with a binary operation that is commutative.

xi Group extension

We say that a group K is a subgroup of another group L and that L is a group extension of K iff $K \subseteq L$ and that the operation of K is that of L restricted to K.

A trivial subgroup of a group is the singleton of the identity element of its operation.

A proper subgroup of a group is a subgroup of it that is a proper subset of it.

xii One-parameter subgroup

A map $\phi : \mathbb{R} \to G$ to some group equipped with a binary operation $*$ is called a one-parameter subgroup of G iff

$$\phi(t) * \phi(s) = \phi(t + s).$$

V Ring-like algebraic structure

TODO rngs ⊃ rings ⊃ commutative rings ⊃ integral domains ⊃ integrally closed domains ⊃ GCD domains ⊃ unique factorization domains ⊃ principal ideal domains ⊃ Euclidean domains ⊃ fields ⊃ algebraically closed fields

i Rng or non-unital ring

A rng or non-unital ring is a ordered triple $(R, +, \cdot)$ of set R, binary operations $+$, and $\cdot$ on R called addition and multiplication such that

- $(R, +)$ is an abelian group under addition with 0 as the identity element, called additive identity,
- $(R, \cdot)$ is a semigroup under multiplication, and

- Distributivity of multiplication over addition:

$$a \cdot (b + c) = (a \cdot b) + (b \cdot c).$$

ii Ring

A ring is a ordered triple $(R, +, \cdot)$ of set R, binary operations $+$, and $\cdot$ on R called addition and multiplication such that

- $(R, +)$ is an abelian group under addition with 0 as the identity element, called additive identity,
- $(R, \cdot)$ is a monoid under multiplication with 1 as the identity element, called multiplicative identity, and
- Distributivity of multiplication over addition:

$$a \cdot (b + c) = (a \cdot b) + (b \cdot c).$$

iii Exponentiation to positive integer power

Exponentiation of an element x of a rng R to a positive integer power n is defined as

$$x^n = \prod_{i=1}^{n} x.$$

iv Exponentiation to zero

Exponentiation of an element x of a ring R to 0 is defined as 1.

v Ring homomorphism

A ring homomorphism is a function from a ring R to a ring S that preserves addition, multiplication and multiplicative identity, that is,

$$f(a + b) = f(a) + f(b), \quad f(ab) = f(a)f(b), \quad f(1) = 1,$$

for all $a, b \in R$.

vi Ring extension

We say that a ring K is a subring of another ring L and that L is a ring extension of K iff $K \subseteq L$ and that the operations of K are those of L restricted to K.

vii Commutative ring

A commutative ring is a ring whose multiplication is commutative.

viii Center of ring

The center of a ring R, denoted as $Z(R)$, is a commutative ring defined as a subring of R consistig of all elements x such that $xy = yx$ for all elements y in R.

ix Module

A left module M of a ring R, denoted as left R-module, is an abelian group $(M, +)$ and an operation $\cdot : R \times M \to M$ called scalar multiplication such that for all $r, s \in R$ and $x, y \in M$,

$$r \cdot (x + y) = r \cdot x + r \cdot y,$$

$$(r + s) \cdot x = r \cdot x + s \cdot x,$$

$$(r \cdot s) \cdot x = r \cdot (s \cdot x),$$

$$1 \cdot x = x.$$

A right module M of a ring R, denoted as right R-module, is an abelian group $(M, +)$ and an operation $\cdot : R \times M \to M$ called scalar multiplication such that for all $r, s \in R$ and $x, y \in M$,

$$r \cdot (x + y) = r \cdot x + r \cdot y,$$

$$(r + s) \cdot x = r \cdot x + s \cdot x,$$

$$(r \cdot s) \cdot x = s \cdot (r \cdot x),$$

$$1 \cdot x = x.$$

If R is a commutative ring, then left R-module is equivalent to right R-module, and they are collectively called R-module.

x Associative algebra

An associative algebra A over a commutative ring R, denoted as associative R-algebra or R-algebra, is a R-module A of which $(A, +)$ is a ring and the scalar multiplication is such that for all $r \in R$ and $x, y \in A$,

$$r \cdot (x \cdot y) = (r \cdot x) \cdot y = x \cdot (r \cdot y).$$

xi Commutative algebra

A commutative algebra is an associative algebra that is also a commutative ring.

xii Algebraic element or algebraic

If A is an associative algebra over a ring K, then an element a of A is an algebraic element or algebraic over K, if there exists some non-zero polynomial $g(x) \in K[x]$ such that $g(a) = 0$. Elements of A that are not algebraic over K are transcendental over K.

xiii Field (域)

A field $(F, +, \cdot)$ is a commutative ring with $1 \neq 0$ and $(F \setminus \{0\}, \cdot)$ being an abelian group.

xiv Field extension

We say that a field K is a subfield of another field L and that L is a field extension of K iff $K \subseteq L$ and that the operations of K are those of L restricted to K.

xv Algebraic extension

An algebraic extension L of a field K is a field extension of K that is an associative algebra over K such that for all $x \in L$, x is algebraic over K. TODO Polynomial polynomial with coefficient in ring R. Root Polynomial ring TODO

xvi Algebraically closed field

A field F is closed if every non-constant polynomial $p(x) \in F[x]$ has a root in F. TODO

xvii Algebraic closure

An algebraic closure of a field K is an algebraic extension of K that is algebraically closed. TODO Separable extension Separable polynomial TODO

xviii Finite field or Galois field

TODO